

LIU Yufeng

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Education Background

Beijing Normal University(BNU)

09/2023-06/2027

- Program: Geographic Information Science (Bachelor's Degree)
- GPA: 78/100, 3.0/4.0

English Language Proficiency

IELTS: 7.0 (L 8.5 R 7.0 S 6.0 W 6.0)

07/03/2026

Publications

- LIU, Yufeng (1st Author); LIU, Suhong (Supervisor). Urban-Scale Chikungunya Risk Mapping in the Western Guangdong-Hong Kong-Macao Greater Bay Area Using Remote Sensing. *International Journal of Environmental Research and Public Health*. Publication Date: 30/05/2026
- LIU, Yufeng (4th Author); LIU, Shiliang (Supervisor). Driving Mechanisms of Spatio-Temporal Vegetation Dynamics in a Typical Agro-Pastoral Transitional Zone in Fengning County, North China. *Land*. Publication Date: 09/01/2026

Honors & Awards

- Honorable Mention in Mathematical Contest In Modeling by COMAP 09/05/2026
- University-level Completion of BNU Undergraduate Innovation and Entrepreneurship Training Program 20/05/2026

Research Projects

Individual Tree Crown Segmentation Project

Supervisor: Prof. Yan Guangjian (BNU) | Project Member | 10/03/2026- Present

- Contribute to the end-to-end development of an individual tree crown instance segmentation pipeline built upon YOLOv8-seg and YOLOv11 architectures;
- Lead dataset processing and code implementation tasks, including format conversion of International Society for Photogrammetry and Remote Sensing benchmark datasets, design of data augmentation pipelines, and iterative hyperparameter tuning for the segmentation model;
- Implemented post-processing workflows (including noise filtering and crown boundary smoothing) to refine segmentation outputs, elevating the model's segmentation accuracy in dense forest and complex urban forest scenarios.

Risk Assessment of Chikungunya Transmission Based on High-resolution Remote Sensing

Supervisor: Prof. Liu Suhong (BNU) | Independent Project Leader | 10/02/2026- 20/05/2026

- Independently oversaw the end-to-end project lifecycle, covering algorithm coding, remote sensing data processing, and academic manuscript preparation, and published one research paper;
- Ingested and fused multi-source Sentinel-2 and Landsat imagery via the Google Earth Engine cloud platform, and employed the Random Forest algorithm to upscale environmental covariate layers to 10-meter spatial resolution;
- Constructed a spatio-temporal Poisson regression model and Mosquito Habitat Suitability Index framework to improve transmission risk assessment accuracy;
- Performed comparative validation against Ecological Niche Modeling outputs to evaluate model performance, and quantified/ranked the key environmental and anthropogenic drivers of chikungunya transmission.

Research on Machine Learning-based Prediction of Typhoon-induced Building Damage

Supervisor: Prof. Tang Hong (BNU) | Project Leader | 10/03/2026- Present

- Lead the end-to-end project lifecycle, including algorithm coding, geospatial data processing, and academic manuscript drafting;
- Integrated the NOAA IBTrACS (International Best Track Archive for Climate Stewardship) tropical cyclone best-track dataset with Maxar commercial high-resolution remote sensing imagery to construct a fine-grained building damage benchmark dataset comprising 701 labeled samples, plus a regional-scale extended dataset covering more than 20,000 buildings; extracted multi-dimensional predictive

features including typhoon wind speed, wind exposure duration, and building structural attributes for model training.

- Conducted comparative performance evaluation of mainstream machine learning algorithms including Random Forest, Multilayer Perceptron, and XGBoost; developed a weighted ensemble learning framework based on soft voting to enhance model stability and generalization capability, and built a probabilistic prediction model for typhoon-induced building damage.
- The paper has been submitted to the International Conference on Remote Sensing and Geoinformation, and is intended for submission to the *International Journal of Disaster Risk Reduction*.

Global Greening Research Review and Bibliometric Analysis Based on Remote Sensing Methods

Supervisor: Assoc. Prof. Yan Kai | Project Leader | 10/03/2026- Present

- Independently managed full-cycle project tasks including data processing, code development, and the paper writing, based on a Web of Science dataset covering 1980- 2025 literature;
- Conducted bibliometric visualization via VOSviewer and CiteSpace to extract field development trends and research hotspots, and built a systematic review framework to analyze core drivers of global greening (climate change, CO₂ fertilization, remote sensing observations);
- The paper is scheduled for submission to *International Journal of Applied Earth Observation and Geoinformation*.

East Africa Ecological Quality Change: Robust Component-based and Spatiotemporal Separated Attribution Analysis

Supervisor: Prof. Liu Shiliang | Project Leader | 10/03/2026- Present

- Coordinated the full project workflow, including data processing, code development, and manuscript preparation. Using East African vegetation cover fraction, standardized leaf area index, and standardized gross primary productivity from 2001 to 2020, developed a novel vegetation ecological quality index.
- Conducted FDR-corrected trend analysis to reveal pronounced spatial heterogeneity in ecological quality, with significantly improved and degraded areas accounting for 21.25% and 3.72% of the region, respectively. The study was refined in collaboration with co-authors from the U.S. and East Africa and is intended for submission to *Communications Earth & Environment*.

Street Tree Suitability Assessment in Xiong'an New Area Based on Multi-source Data Fusion and Comprehensive Evaluation Model

Supervisor: Assoc. Prof. Yan Kai | Project Leader | 10/03/2026- Present

- Lead the end-to-end project lifecycle, encompassing research conceptualization, experimental design, data processing, model implementation, and final research output delivery.
- Leveraged multi-source Earth observation data, integrating Landsat and Sentinel optical remote sensing imagery with ERA5 meteorological reanalysis data to retrieve core ecological and environmental variables through inversion, deriving three categories of key assessment indicators: vegetation dynamics, surface thermal environment, and ambient atmospheric pollutants.
- The project is supported by a university-level research grant from BNU and has been formally concluded. All research outputs have been compiled into a peer-reviewed academic manuscript, which is currently under review following submission to the 3rd Interdisciplinary Forum on Artificial Intelligence and Remote Sensing.

Extracurricular Activities

- Student Union, Faculty of Geographical Science, BNU 07/ 2024- 07/2025
Deputy Head of Publicity Department
- Symphonic Wind Orchestra, BNU 09/2023- Present
Clarinet Section Leader

Computer Skills

- Programming Languages: Python, C++, MATLAB, Java
- GIS & Remote Sensing: ArcGIS, QGIS, ENVI, Google Earth Engine
- Data Analysis & Statistics: SPSS, GeoDa